TASK – 3

Maximum Profit Analysis Java

PROGRAM :

```
import java.util.*;

public class MaximumProfit {
   public static void main(String[] args) {

      List<Integer> prices = Arrays.asList(
         7, 1, 5, 3, 6, 4
      );

      int minPrice = prices.get(0);
      int maxProfit = 0;

      for (int price : prices.stream().toList()) {
         minPrice = Math.min(minPrice, price);
         maxProfit = Math.max(maxProfit, price - minPrice);
      }

      System.out.println("Stock Prices: " + prices);
      System.out.println("Maximum Profit: " + maxProfit);
   }
}
```

INPUT :

Stock Prices: [7, 1, 5, 3, 6, 4]

OUTPUT :

Stock Prices: [7, 1, 5, 3, 6, 4]

Maximum Profit: 5